Tanmay Pandey

5^{th} Year BSMS, Indian Institute of Science Education and Research (IISER) Mohali
MS Thesis Internship, Max Planck Institute for Biophysics, Germany
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EDUCATION

2022–2027
BS–MS Dual Degree in Biology
Physics Minor
Indian Institute of Science Education and Research (IISER) Mohali

SKILLS

- **Microscopy**
  Fluorescence, phase-contrast, and time-lapse imaging
- **Wet-lab**
  GPMV isolation and animal cell culture
- **Image analysis**
  CNN segmentation, contour tracking, curvature and fluctuation analysis
- **Computation**
  Python, MATLAB, statistics, and ML-assisted pipelines
- **Software**
  ImageJ (Fiji), Chimera

ADMINISTRATIVE ROLES

2025–2026
General Secretary & Hostel Representative
Student Representative Council (SRC)
IISER Mohali

RESEARCH PROFILE

Final year BS–MS student majoring in Biology with a minor in Physics at IISER Mohali, working on membrane mechanics, plasma membrane vesicles, and quantitative image analysis. Current work involves quantitative analysis of giant plasma membrane vesicles (GPMVs) using computational tools on microscopy data to uncover the physics behind cell death.

RESEARCH EXPERIENCE

Micromechanics of Membrane Blebs in Regulated Necrotic Cell Death **May 2026 – Ongoing**
MS Thesis Project – Soft Matter Biophysics Lab, IISER Mohali (India) & Dept. of Membrane Dynamics, MPI Biophysics (Germany)
Co-supervisors: Dr. Uris Ros, Dr. Tripta Bhatia
Investigating membrane mechanical remodeling during regulated cell death using GPMVs, measuring rigidity, tension, permeability, curvature, and shape fluctuations through microscopy and fluctuation spectroscopy. Studying pathway-specific mechanical signatures of cell death using image-analysis pipelines.

Membrane Mechanics and Plasma Membrane Vesicles **Apr 2023 – Apr 2026**
Project Intern – Soft Matter Biophysics Lab, IISER Mohali
Supervisor: Dr. Tripta Bhatia
Quantitatively analyzed GPMV and GUV morphology to extract curvature, reduced volume, and membrane mechanical parameters; developed Python pipelines for contour detection and morphological analysis. Performed GUV electroformation, GPMV preparation, and multi-channel fluorescence microscopy.

DNA-Based Physical Reservoir Computing Systems **Mar 2024 – Dec 2025**
Student Assistant (Remote) – Bio-inspired Computation Lab, University of Kiel
Supervisor: Prof. Jan Steinkühler
Developed numerical models of DNA-based spring–mass systems as physical reservoirs inspired by mass-spring dynamics, and analyzed the adaptability and performance of these reservoir networks across nonlinear dynamical tasks.

PUBLICATIONS

2026 Quantitative Analysis of Shape Changes in Lipid Vesicles Using Deep Learning-Based Image Segmentation and Tracking Methods
T. Pandey, R. Kudawla, J. V. B. S., C. Devarakonda, A. Chatterjee, T. Bhatia. *Under revision.*

2026 Directed evolution effectively selects for DNA-based physical reservoir computing networks
T. Pandey, P. Feketa, J. Steinkühler. Neuromorphic Computing and Engineering (IOP). DOI: 10.1088/2634-4386/ae2cc3

2025 Experimentally determined shapes of plasma membrane vesicles and PC/PC–cholesterol vesicles
H. Kaur, R. Kudawla, *T. Pandey*, T. Bhatia. Journal of Physical Chemistry B (ACS). DOI: 10.1021/acs.jpcb.4c07431

2024 Shape analysis of biomimetic and plasma membrane vesicles

R. Kudawla, H. Kaur, *T. Pandey*, T. Bhatia. ChemSystemsChem (Wiley). DOI: 10.1002/syst.202400052

RESEARCH SOFTWARE

2025 Vesicle Shape Analysis Toolkit

T. Pandey, T. Bhatia. Open-source software for contour extraction, curvature, and mechanical analysis of vesicle membranes (Zenodo + GitHub). DOI: 10.5281/zenodo.17104430